The chain

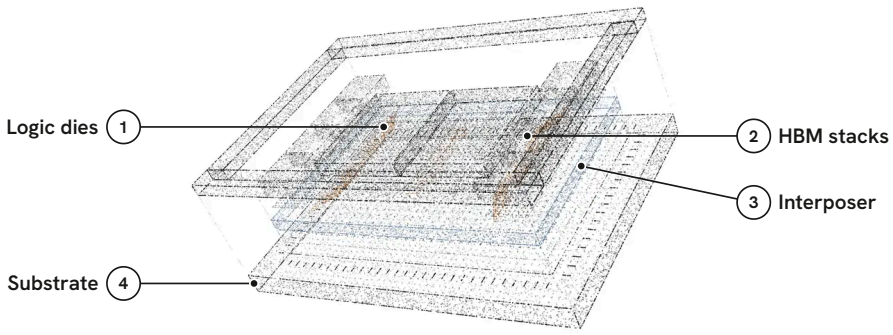
01 Chip Design, EDA and IP
02 Silicon and Wafers
03 Chemicals, Gases and Photoresist
04 Lithography
05 Photomasks and Pellicles
06 Deposition and Etch
07 Metrology and Inspection
08 Foundries and Fabs
09 Memory and HBM
10 **Advanced Packaging**
11 Substrates and PCBs
12 Test and Assembly
13 Systems and Networking
14 Data Centers and Power

15 The Economics of AI Chips
16 Who Controls the Chain

Stage 10 of 14 · Extreme concentration · Hard to substitute

Advanced Packaging

Epoch AI estimates that in 2025 Nvidia, Google, AMD and Amazon used around 90 percent of the capacity for CoWoS, TSMC's packaging for AI chips.



Specimen 10 · Advanced Packaging

A CoWoS-L package, pulled apart. Two logic dies and eight HBM stacks sit on the interposer, shown in silver with its copper wiring. In CoWoS-L the interposer is fine redistribution wiring with small embedded silicon bridges. It sits on a resin substrate with a grid of solder balls underneath. The chips together cover about 2,700 mm², a little over three reticle fields. Footprints are to scale. Thicknesses and the gaps between layers are exaggerated.

Parts, as numbered on the drawing

1. Logic dies, two: see stage 08, Foundries and Fabs
2. HBM stacks, eight: see stage 09, Memory and HBM
3. Interposer: TSMC (TW), as CoWoS
4. Substrate: see stage 11, Substrates and PCBs

In plain terms

Packaging joins finished chips into one component and mounts them on a small circuit board, called a substrate, wired far more coarsely than the chips themselves. A lithography machine, which prints a chip's circuits, can cover only a rectangle about the size of a postage stamp in one shot. Many leading accelerators, Nvidia's Blackwell among them, need more logic than fits in that rectangle. So they are built from several chips, called dies, set side by side on an interposer, a layer of much finer wiring that links them and sits on the substrate.

Stacks of memory chips sit on the same interposer, a few millimeters away. The interposer is either one piece of silicon or layers of fine copper wiring that often carry small silicon bridges where the links are densest. Silicon expands less than the resin substrate beneath it when heated, so the package can warp and its joints crack. Preventing that in high volume is hard. Nvidia, Google, AMD and Amazon all use TSMC's interposer packaging, called CoWoS, made in Taiwan.

In short

Epoch AI estimates that in 2025 Nvidia used 60.3 percent of all capacity for CoWoS, TSMC's interposer packaging[1]. On the July 2026 earnings call, TSMC's chief executive, C.C. Wei, said its packaging capacity was so tight that it limited customers' growth[2]. Asked about Intel's EMIB-T packaging, he said he hoped it would take some of the load off TSMC[2].

In a TrendForce ranking by 2024 revenue, which CSET cites, four of the top ten outsourced firms that package and test chips are headquartered in China[3]. But in 2025 Chinese firms had 5.3 percent of the market for advanced packaging fabrication tools, against 65.8 percent for US firms, by CSET's analysis of TechInsights data[4].

A lithography scanner prints a rectangle 26 mm by 33 mm, or 858 mm^2, in one exposure[5]. That area is called the reticle field. Every leading AI accelerator, counting its memory stacks, covers more area than that. Packaging is the stage that lets an accelerator be larger than one reticle field, and the packaging method limits how much larger it can be.

Chokepoint card Advanced Packaging

Concentration

Extreme

Three or fewer firms could supply this stage today, or one firm holds more than half of it.

Substitutability

Hard

Rated on CoWoS capacity, which TSMC said in July 2026 was so tight that it limited its customers' growth[2].

Price or market size

About $1,100

for the CoWoS-L packaging of one Nvidia B200, plus about $1,000 for the dies and memory lost in failed packages, against about $900 for its two logic dies, by Epoch AI's December 2025 estimate[6]

1 Advanced packaging and HBM, not logic dies, were the bottlenecks on AI chip... Epoch AI, 12 March 2026. epoch.ai
2 Q2 2026 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... LSEG StreetEvents, 16 July 2026, p. 10. investor.tsmc.com
3 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu
4 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 20 October 2026. chipexplorer.eto.tech
5 TWINSCAN NXE:3400C. ASML, accessed 9 September 2026. asml.com
6 NVIDIA's B200 costs around $6,400 to produce. Epoch AI, 10 December 2025. epoch.ai

Who leads	TW	**TSMC**	CoWoS packaging used by Nvidia, Google, AMD and Amazon in 2025[1]
	US	**Intel**	Foveros 3D packaging in volume at Rio Rancho, New Mexico, since January 2024[7]
	TW	**ASE Technology**	NT$389.2 billion assembly, test and materials revenue, 2025[8]
	US	**Amkor**	$6.71 billion net sales, 2025[9]
	CN	**JCET**	RMB 38.87 billion revenue, 2025[10]
	SG	**ASMPT**	Thermocompression bonders, with $532.1 million of advanced packaging revenue in 2025[11]
Where it is made	TW	**Taiwan**	TSMC advanced packaging fabs and a cluster of outsourced assembly and test firms
	US	**United States**	Intel Foveros packaging in volume at Rio Rancho, New Mexico
	CN	**China**	Four of the top ten outsourced assembly and test firms by 2024 revenue are Chinese, in a TrendForce ranking cited by CSET
	NL	**Netherlands**	Besi hybrid bonders
	JP	**Japan**	Disco grinders and dicers
Why substitution is slow			CoWoS packaging has been outsourced to assembly firms such as ASE and Amkor before[12]. Intel's EMIB, which links chips through silicon bridges embedded in the substrate, is another option, and Amazon's Trainium and Google's TPU were both adopting it as of March 2026[12]. But a newcomer would have to match TSMC, which said in April 2026 that it was producing CoWoS at 5.5 times the area a scanner prints in one shot[13].

Where China stands

Chinese firms had 5.3 percent of the market for fabrication tools built for advanced packaging in 2025, by CSET's analysis of TechInsights data[4]. Between 2019 and 2024 they gained 5 percentage points of share in test tools but lost share in bonding and other conventional assembly and packaging tools[3].

Where the US stands

Intel opened Fab 9 in Rio Rancho, New Mexico, in January 2024 to make its Foveros 3D packages in volume[7]. Amkor broke ground in October 2025 on a $7 billion packaging and test campus in Peoria, Arizona, and expects production to start in early 2028[14].

01

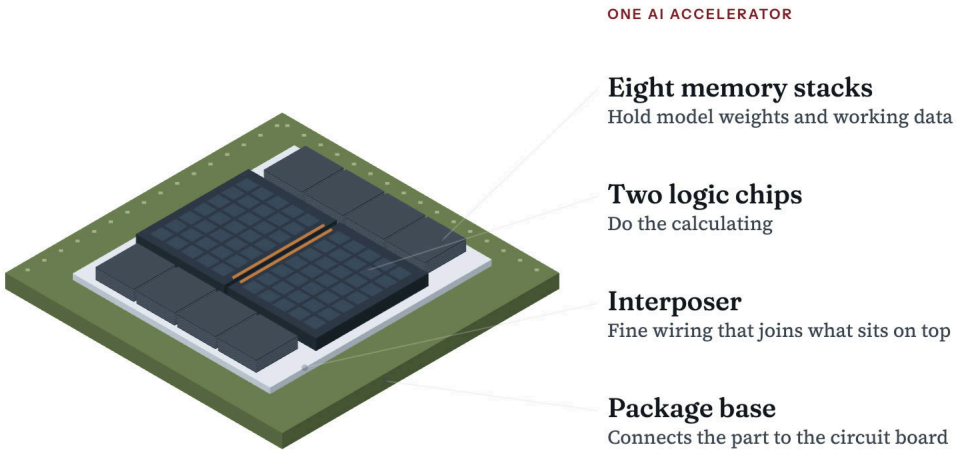
How advanced packaging and CoWoS work

A second reason for the interposer is high bandwidth memory (HBM). An HBM4 stack connects through 2,048 data signals, packed too tightly for a standard resin substrate, so the memory sits beside the logic on the interposer[15]. Even Nvidia's H100, whose single logic die is 814 mm² and fits inside one reticle field, carries five HBM stacks[16]. Epoch AI counts TSMC's CoWoS packaging among the three inputs the H100 needs, along with its logic die and its HBM[1].

Figure 10.1 · Explainer, 19 seconds

1 Advanced packaging and HBM, not logic dies, were the bottlenecks on AI chip... Epoch AI, 12 March 2026. epoch.ai
3 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu
4 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 20 October 2026. chipexplorer.eto.tech
7 Intel Opens Fab 9 in New Mexico. Intel, 24 January 2024. intel.com
8 ASE Technology Holding Co., Ltd. Reports Its Unaudited Consolidated Financial... PR Newswire, 5 February 2026. prnewswire.com
9 Amkor Technology, current report on Form 8-K dated 9 February 2026. U.S. Securities and Exchange Commission, 9 February 2026. sec.gov
10 JCET Releases 2025 Annual Report, Posts Record-High Full-Year Revenue and... JCET Group, 9 April 2026. jcetglobal.com
11 ASMPT Announces 2025 Annual Results AI-Driven Structural Growth Underpins Group... ASMPT, 4 March 2026. asmpt.com
12 The Great AI Silicon Shortage. SemiAnalysis, 12 March 2026. newsletter.semianalysis.com
13 TSMC Debuts A13 Technology at 2026 North America Technology Symposium. TSMC, 23 April 2026. pr.tsmc.com
14 Amkor Technology Breaks Ground on New Semiconductor Advanced Packaging and Test... Amkor Technology, 6 October 2025. ir.amkor.com
15 New JEDEC® SPHBM4 Standard Enables HBM4-Class Bandwidth on Organic Substrates. JEDEC Solid State Technology Association, 13 July 2026. jedec.org
16 NVIDIA Hopper Architecture In-Depth. Nvidia, 22 March 2022. developer.nvidia.com

The parts of an AI accelerator



1. **Package base.** An AI chip is built from several parts. First comes the package base, a layered board that connects everything to the circuit board below.
2. **Interposer.** On it sits an interposer, a thin layer of very fine wiring that joins everything placed on top.
3. **Logic chips.** Two logic chips go in the middle. They do the calculating, and each is as large as a chipmaking machine can print.
4. **Memory.** Eight stacks of memory go beside them, a few millimeters away. They hold the AI model's weights and the record of each conversation so far.
5. **One package.** Together they make one AI accelerator. Most are put together by TSMC, in Taiwan.

Note: Simplified view of a two-die, eight-stack accelerator. Footprints follow the sizes given in this chapter and chapter 9. Heights are exaggerated so each layer shows.

CoWoS stands for chip on wafer on substrate, the order in which the parts are joined. For CoWoS-S, the version whose interposer is one piece of silicon, TSMC makes a silicon wafer that carries wiring and vertical copper vias but no transistors. Vias are metal-filled holes that join one layer of wiring to the next.

TSMC bonds the logic dies and memory stacks face down onto that wafer with microbumps. These are dots of solder, and each dot is one electrical joint. In Intel's Foveros 2.5D packaging, which can also set dies side by side on a silicon interposer, the microbumps sit 36 micrometers apart, less than half a hair's width[17]. The spacing differs by design, and SK hynix gives about 20 micrometers as typical for microbumps[18]. Then TSMC cuts the interposer out of the wafer and attaches it to the resin substrate, which later goes onto a circuit board.

One cracked joint can scrap a package holding eight tested HBM stacks and two large, expensive logic dies. Epoch AI estimated in December 2025 that CoWoS-L packaging, the version built with small silicon bridges, added about $1,100 to the cost of building one Nvidia B200[6]. That is more than its estimate of about $900 for the B200's two logic dies[6]. Epoch AI adds about $1,000 more to each B200 for the packages that fail testing, whose logic dies and memory stacks are lost[6].

6 NVIDIA's B200 costs around $6,400 to produce. Epoch AI, 10 December 2025. epoch.ai
17 Foveros Direct 3D Technology Brief. Intel, 6 February 2026. intel.com
18 [Tech Note] Hybrid Bonding: Evolving into a Foundational Technology for... SK hynix, 25 August 2026. news.skhynix.com

02

Types of advanced packaging

Family	Carrier	Size limit	Used for
CoWoS-S	Full silicon interposer	Up to 3.3× reticle	H100-class parts
CoWoS-R	Resin interposer with fine redistribution wiring	Above 3.3× reticle	Cost-sensitive, fewer HBM stacks
CoWoS-L	Redistribution wiring with local silicon bridges	5.5× reticle in production, 9.5× in development	Blackwell

2.5D: silicon, RDL and bridges

In 2.5D packaging the chips sit side by side and connect through an interposer or silicon bridges beneath them. TSMC has run CoWoS in volume since 2012[19]. CoWoS-S has the finest wiring, but its interposer is one unbroken piece of silicon whose yield (the share of interposers that work) falls as the area grows. CoWoS-L replaces that piece of silicon with redistribution layers (RDL) of fine copper wiring and embeds small silicon bridges where the wiring is densest[20].

Intel's EMIB embeds similar silicon bridges in the resin substrate[17]. SemiAnalysis reported in March 2026 that EMIB was gaining traction, with Amazon's Trainium and Google's TPU both adopting it[12].

Blackwell was the first high-volume design on CoWoS-L, and packaging problems at TSMC and in Nvidia's design limited its supply[21]. The bridges that carry the 10 TB/s link between the two logic dies must be placed very precisely[21]. The dies, bridges, interposer and substrate also expand at different rates, and that mismatch caused warpage[21].

3D: SoIC and Foveros

In 3D packaging one die sits on top of another, and the joints between them are moving from solder microbumps to hybrid bonding. Intel's Foveros stacks dies face to face and joins them with solder microbumps[17]. Its newer Foveros Direct drops the solder and bonds the copper pads directly, a method called hybrid bonding[17]. Hybrid bonding joins the insulating layer around the pads as well as the copper, with no bumps in between[18]. That lets the connections sit far closer together than microbumps allow[17]. TSMC's SoIC also uses hybrid bonding, and AMD's 3D V-Cache processors use SoIC to attach an extra cache die to the processor die[22]. Stacking also changes how power reaches the top die, which has to draw it up through vertical wires in the die below, called through-silicon vias (TSVs)[22].

12 The Great AI Silicon Shortage. SemiAnalysis, 12 March 2026. newsletter.semianalysis.com

17 Foveros Direct 3D Technology Brief. Intel, 6 February 2026. intel.com

18 [Tech Note] Hybrid Bonding: Evolving into a Foundational Technology for... SK hynix, 25 August 2026. news.skhynix.com

19 TSMC, CoWoS technology page. TSMC, accessed 9 September 2026. 3dfabric.tsmc.com

20 TSMC 2025 Annual Report, chapter 5. TSMC, 16 April 2026, p. 3. investor.tsmc.com

21 Nvidia's Blackwell Reworked - Shipment Delays & GB200A Reworked Platforms. SemiAnalysis, 4 August 2024. semianalysis.com

22 Hybrid Bonding Process Flow - Advanced Packaging Part 5. SemiAnalysis, 9 February 2024. newsletter.semianalysis.com

TSMC put SoIC into volume production in 2022, stacking SRAM memory chips on logic wafers[23]. Its 3 nm SoIC entered volume production in 2025[20]. An A14-on-A14 version, stacking dies made on TSMC's 1.4 nm process, is due in 2029[13]. It will carry 1.8 times the die-to-die connection density of SoIC that stacks 2 nm dies on 2 nm dies[13].

SoIC bonding also brings optics into the package. TSMC's COUPE uses SoIC to join a silicon photonics chip, which sends data as light, to its electrical control chip[20]. In April 2026 TSMC planned to begin producing COUPE on substrate later that year[13]. It also said that putting COUPE inside the package gives twice the power efficiency of a pluggable version on the circuit board[13].

Two ways off the silicon interposer

Panel-level packaging builds packages on a large rectangular panel instead of a round wafer, so rectangular packages fit it with less waste at the edges. See Test and assembly. JEDEC, the semiconductor standards body, published a standard called SPHBM4 on 13 July 2026. It puts HBM4 memory on a base chip carrying 512 data signals instead of 2,048, each running four times as fast to keep the same bandwidth[15]. With fewer signals, the connections can be spaced widely enough to mount the stack on a standard resin substrate instead of the silicon interposer[15].

03

TSMC's CoWoS capacity and its buyers

TSMC does not publish its CoWoS capacity[1]. Industry estimates from SemiWiki and TrendForce, compiled by Epoch AI, put it at about 65,000 to 75,000 wafers a month at the end of 2025, up from 35,000 to 40,000 a year earlier[1].

On the July 2024 earnings call, TSMC's chief executive, C.C. Wei, said TSMC was more than doubling CoWoS capacity in 2024 and would more than double it again in 2025[24]. He said he could not yet balance supply and demand[24]. At its 2024 symposium TSMC had said CoWoS capacity would grow at a 60 percent compound annual rate over the next few years[24]. Asked about that rate on the call, Wei did not repeat it[24].

In October 2025 Wei would say only that TSMC was still working to increase CoWoS capacity for 2026, and he gave no figure[25]. He said front-end capacity (making the chips) and back-end capacity (packaging them) were both, in his words, very tight[25].

By Epoch AI's estimate, four chip designers used most of the world's CoWoS capacity in 2025 but only a small share of its advanced chipmaking. Epoch AI compared each chip designer's use of components with world supply[1]. It puts Nvidia at 60.3 percent of all CoWoS capacity in 2025, with a 90 percent confidence interval of 56.5 to 64.3[1]. Google took 13.5 percent, AMD 8.4 and Amazon 7.4[1]. The four also used around 90 percent of

1 Advanced packaging and HBM, not logic dies, were the bottlenecks on AI chip... Epoch AI, 12 March 2026. epoch.ai

13 TSMC Debuts A13 Technology at 2026 North America Technology Symposium. TSMC, 23 April 2026. pr.tsmc.com

15 New JEDEC® SPHBM4 Standard Enables HBM4-Class Bandwidth on Organic Substrates. JEDEC Solid State Technology Association, 13 July 2026. jedec.org

20 TSMC 2025 Annual Report, chapter 5. TSMC, 16 April 2026, p. 3. investor.tsmc.com

23 TSMC 2022 Annual Report, page 97. TSMC, accessed 18 October 2026. investor.tsmc.com

24 Q2 2024 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... Refinitiv StreetEvents, 18 July 2024, p. 6. investor.tsmc.com

25 Q3 2025 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... Refinitiv StreetEvents, 16 October 2025, p. 7. investor.tsmc.com

global HBM supply in 2025, measured by value[1]. But they used only 12 percent of advanced logic die production[1].

Figure 10.2

Nvidia used an estimated 60.3 percent of CoWoS capacity in 2025

Estimated share of CoWoS capacity used, 2025



Source: Epoch AI, advanced packaging and HBM were the bottlenecks on AI chip production in 2025 [1]

In March 2026 the research firm SemiAnalysis judged that front-end capacity had become the dominant bottleneck and that CoWoS constraints were easing[12]. It said TSMC was planning CoWoS capacity around how many wafers N3, its 3 nm process, could produce, because added packaging capacity is of no use without front-end wafers to fill it[12].

TSMC gave a different account four months later. On the July 2026 earnings call, Wei said back-end capacity was still in shortage[26]. Asked about Intel's EMIB-T, he said he welcomed the extra flexibility and hoped it would take some of the load off TSMC[2].

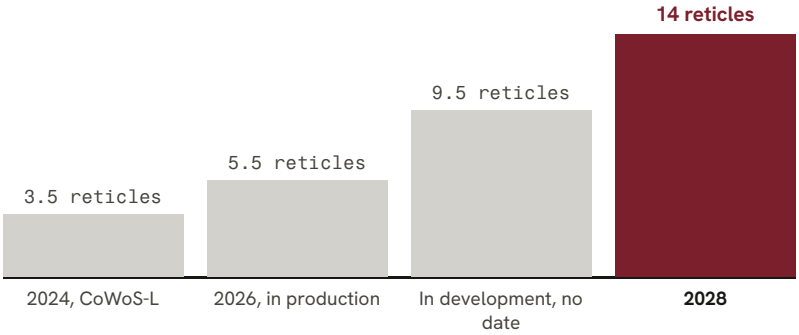
04

Who makes advanced packages

Figure 10.3

TSMC plans 14-reticle CoWoS interposers for 2028, up from 3.5 in 2024

CoWoS interposer size on TSMC's roadmap



Source: TSMC 2025 annual report and 2026 technology symposium [13]

Note: The 2024 column is CoWoS-L. CoWoS-S, whose interposer is one piece of silicon, reaches up to 3.3 reticles.

1 Advanced packaging and HBM, not logic dies, were the bottlenecks on AI chip... Epoch AI, 12 March 2026. epoch.ai

2 Q2 2026 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... LSEG StreetEvents, 16 July 2026, p. 10. investor.tsmc.com

12 The Great AI Silicon Shortage. SemiAnalysis, 12 March 2026. newsletter.semianalysis.com

13 TSMC Debuts A13 Technology at 2026 North America Technology Symposium. TSMC, 23 April 2026. pr.tsmc.com

26 Q2 2026 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... LSEG StreetEvents, 16 July 2026, p. 17. investor.tsmc.com

Three kinds of firm package chips. TSMC, which makes chips for others, and Intel, which makes its own, do it in their own back-end fabs. Outsourced assembly and test firms do the rest. Intel opened Fab 9 in Rio Rancho, New Mexico, in January 2024 to make its Foveros 3D packages in volume[7]. It called the plant the only US factory producing the most advanced packaging at scale[7].

ASE leads the outsourced assembly and test firms, and Amkor is the largest one headquartered in the United States. ASE Technology Holding had revenue of NT$645.4 billion in 2025, about 40 percent of it from its electronics manufacturing arm, a business separate from assembly and test[8]. Its assembly, test and materials business brought in NT$389.2 billion, up from NT$325.9 billion in 2024[8]. Amkor had net sales of $6.71 billion in 2025[9].

Each of the key packaging tools comes from a small number of makers.

- **Thermocompression bonding.** Pressing dies together under heat and force. ASMPT's advanced packaging revenue reached $532.1 million in 2025, with thermocompression bonder revenue up about 146 percent[11]. ASMPT targets 35 to 40 percent of a market it puts at $1.6 billion in 2028[11].
- **Hybrid bonding.** Copper pads and the insulator around them bond directly, with no solder. Applied Materials co-developed its Kinex hybrid bonding system with Besi, which Applied calls the industry leader in hybrid bonding[27].
- **Grinding and dicing.** Disco held 87 percent of the $1.9 billion world market for dicing tools in 2025, by CSET's count from TechInsights data[28]. The category covers the grinders that thin wafers and the saws that cut them into chips[28]. See Test and assembly.

TSMC and Amkor both plan advanced packaging plants in Arizona. TSMC's Arizona commitment reached $265 billion in July 2026 and covers two advanced packaging plants alongside 10 fabs[29]. Amkor started building a $7 billion campus in Peoria, Arizona, in October 2025 and expects production from early 2028[14]. The US Commerce Department awarded Amkor up to $407 million in CHIPS Act funding for the site in December 2024[30]. In October 2025 Wei said a large outsourced packaging partner, which he did not name, was already breaking ground in Arizona[31]. He said its plant was on an earlier schedule than TSMC's own two packaging fabs there[31].

7 Intel Opens Fab 9 in New Mexico. Intel, 24 January 2024. intel.com

8 ASE Technology Holding Co., Ltd. Reports Its Unaudited Consolidated Financial... PR Newswire, 5 February 2026. prnewswire.com

9 Amkor Technology, current report on Form 8-K dated 9 February 2026. U.S. Securities and Exchange Commission, 9 February 2026. sec.gov

11 ASMPT Announces 2025 Annual Results AI-Driven Structural Growth Underpins Group... ASMPT, 4 March 2026. asmpt.com

14 Amkor Technology Breaks Ground on New Semiconductor Advanced Packaging and Test... Amkor Technology, 6 October 2025. ir.amkor.com

27 Applied Materials, Kinex integrated die-to-wafer hybrid bonding system product... Applied Materials, accessed 9 September 2026. appliedmaterials.com

28 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

29 TSMC Announces Additional $100 Billion Investment in Arizona. Arizona Commerce Authority, 16 July 2026. azcommerce.com

30 Amkor Technology, annual report on Form 10-K for the fiscal year ended 31... U.S. Securities and Exchange Commission, 20 February 2026. sec.gov

31 Q3 2025 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... Refinitiv StreetEvents, 16 October 2025, p. 13. investor.tsmc.com

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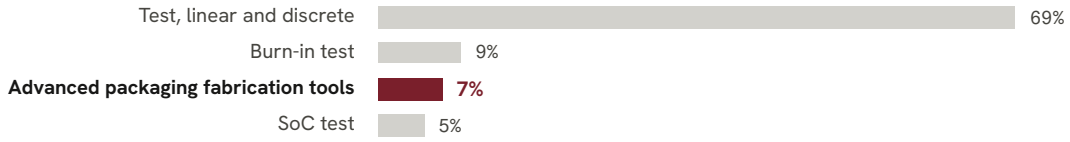
The chokepoint

Nvidia, Google, AMD and Amazon all use TSMC's interposer packaging, called CoWoS, made in Taiwan. By Epoch AI's estimate, those four used around 90 percent of global CoWoS capacity in 2025[1]. In 2024 TSMC was more than doubling CoWoS capacity and still could not keep up with demand[24]. On the July 2026 earnings call, TSMC's chief executive, C.C. Wei, said its packaging capacity was so tight that it limited customers' growth[2]. CoWoS packaging can be, and has been, outsourced to assembly firms such as ASE and Amkor[12].

Figure 10.4

China-based suppliers held 7 percent of the market for advanced packaging fabrication tools in 2024

China-based suppliers' share of advanced packaging fabrication tools and three test tool segments, 2024



Source: CSET, Inside Beijing's Chipmaking Offensive [3]

Four of the top ten outsourced assembly and test firms by 2024 revenue are headquartered in China, by a TrendForce ranking that CSET cites[3]. JCET, a leading Chinese assembly and test firm, had record revenue of RMB 38.87 billion in 2025, up 8.1 percent[10]. JCET says its XDFOI packaging platform has entered high-volume manufacturing with redistribution-layer interposer, silicon interposer and silicon bridge options[32]. Those are the same three kinds of carrier that TSMC sells as CoWoS-R, CoWoS-S and CoWoS-L.

Chinese firms had a small share of the market for advanced packaging tools in 2024 and 2025, and they lost ground in bonding. In 2024 Chinese firms had 7 percent of the market for advanced packaging fabrication tools[3]. Between 2019 and 2024, they gained 5 percentage points of share in test tools and 4 points in fabrication tools built for advanced packaging, by CSET's analysis of TechInsights data[3]. But they lost share in bonding and other conventional assembly and packaging tools, the category where CSET counts hybrid and thermocompression bonders[3]. Within the advanced packaging fabrication tools, Chinese firms gained 10 points in etch and clean tools and 3 points in deposition tools[3]. They lost 2 points each in lithography and polishing (CMP) tools[3]. CSET's Chip Explorer, a separate data tool also built on TechInsights data, puts Chinese firms at 5.3 percent of the $6.2 billion world market for advanced

1 Advanced packaging and HBM, not logic dies, were the bottlenecks on AI chip... Epoch AI, 12 March 2026. epoch.ai
2 Q2 2026 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... LSEG StreetEvents, 16 July 2026, p. 10. investor.tsmc.com
3 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu
10 JCET Releases 2025 Annual Report, Posts Record-High Full-Year Revenue and... JCET Group, 9 April 2026. jcetglobal.com
12 The Great AI Silicon Shortage. SemiAnalysis, 12 March 2026. newsletter.semianalysis.com
24 Q2 2024 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... Refinitiv StreetEvents, 18 July 2024, p. 6. investor.tsmc.com
32 JCET Accelerates Strategic Shift Toward High-End Advanced Packaging, 2025... JCET Group, 21 April 2026. jcetglobal.com

packaging fabrication tools in 2025[4]. It puts US firms at 65.8 percent and Japanese firms at 15.3 percent[4].

TSMC said in April 2026 that it was producing CoWoS at 5.5 times the size of a reticle field and planning larger versions[13]. Each generation on TSMC's roadmap needs a larger interposer.

- **2028.** A 14-reticle interposer, about the area of a compact disc, carrying about ten large compute dies and 20 HBM stacks[13].
- **2029.** More than 14 reticles, plus a 40-reticle System-on-Wafer[13]. TSMC describes System-on-Wafer as joining different data center chips at the scale of a whole wafer[33].

Four questions to answer from memory

1. Why is an AI accelerator built from several chips in one package? see How it works
2. Why is packaging hard? see How it works
3. Which company packages the leading AI chips? see How it works
4. Why has packaging limited how many AI chips can be made? see Capacity

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Four things to remember

- Many leading accelerators, Nvidia's Blackwell among them, need more logic than one chip can hold, so packaging joins several chips and their memory.
- Nvidia, Google, AMD and Amazon all rely on TSMC, in Taiwan, to package their accelerators.
- Packaging was one of the main limits on AI chip output in 2025. In July 2026 TSMC said its packaging capacity was still so tight that it limited customers' growth.
- No US site yet packages Nvidia-class accelerators with HBM on CoWoS, though Intel makes its own Foveros 3D packages in volume in New Mexico.

Sources (33)

1. A Advanced packaging and HBM, not logic dies, were the bottlenecks on AI chip production in 2025. Epoch AI · 12 March 2026
   epoch.ai/data-insights/ai-chip-supply-chain-constraints
2. A Q2 2026 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese, English) — edited transcript. LSEG StreetEvents (transcript of a TSMC earnings call), via TSMC · 16 July 2026 · p. 10
   investor.tsmc.com/english/encrypt/files/encrypt_file/reports/2026-08/3e494f0c14dd0890f897aa044415e21d93486cc4/TSMC%202Q26%20Transcript.pdf#page=10
3. A Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology (CSET) · 14 July 2025
   cset.georgetown.edu/article/inside-beijings-chipmaking-offensive/
4. A Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) · Accessed 20 October 2026
   chipexplorer.eto.tech/?parentNode=N69&selectedNode=N109
5. A TWINSCAN NXE:3400C. ASML · Accessed 9 September 2026
   www.asml.com/en/products/euv-lithography-systems/twinscan-nxe3400c
6. A NVIDIA's B200 costs around $6,400 to produce. Epoch AI · 10 December 2025
   epoch.ai/data-insights/b200-cost-breakdown
7. A Intel Opens Fab 9 in New Mexico. Intel · 24 January 2024
   www.intel.com/content/www/us/en/newsroom/news/intel-opens-fab-9-new-mexico.html

4 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 20 October 2026. chipexplorer.eto.tech
13 TSMC Debuts A13 Technology at 2026 North America Technology Symposium. TSMC, 23 April 2026. pr.tsmc.com
33 TSMC 2025 Annual Report, chapter 5. TSMC, 16 April 2026, p. 4. investor.tsmc.com

8. A ASE Technology Holding Co., Ltd. Reports Its Unaudited Consolidated Financial Results for the Fourth Quarter and the Full Year of 2025. PR Newswire · 5 February 2026
www.prnewswire.com/news-releases/ase-technology-holding-co-ltd-reports-its-unaudited-consolidated-financial-results-for-the-fourth-quarter-and-the-full-year-of-2025-302679779.html

9. A Amkor Technology, current report on Form 8-K dated 9 February 2026. U.S. Securities and Exchange Commission (filing by Amkor Technology) · 9 February 2026
www.sec.gov/Archives/edgar/data/1047127/000104712726000007/amkr123125erex-991.htm

10. A JCET Releases 2025 Annual Report, Posts Record-High Full-Year Revenue and Higher Profit Before Tax. JCET Group · 9 April 2026
www.jcetglobal.com/en/site/news-detail?id=2048

11. A ASMPT Announces 2025 Annual Results AI-Driven Structural Growth Underpins Group Performance. ASMPT · 4 March 2026
www.asmpt.com/en/investor-relations/news-events/asmpt-announces-2025-annual-results/

12. B The Great AI Silicon Shortage. SemiAnalysis · 12 March 2026
newsletter.semianalysis.com/p/the-great-ai-silicon-shortage

13. A TSMC Debuts A13 Technology at 2026 North America Technology Symposium. TSMC · 23 April 2026
pr.tsmc.com/english/news/3302

14. A Amkor Technology Breaks Ground on New Semiconductor Advanced Packaging and Test Campus in Arizona; Expands Investment to $7 Billion. Amkor Technology · 6 October 2025
ir.amkor.com/news-releases/news-release-details/amkor-technology-breaks-ground-new-semiconductor-advanced

15. A New JEDEC® SPHBM4 Standard Enables HBM4-Class Bandwidth on Organic Substrates. JEDEC Solid State Technology Association · 13 July 2026
www.jedec.org/news/pressreleases/new-jedec%C2%AE-sphbm4-standard-enables-hbm4-class-bandwidth-organic-substrates

16. A NVIDIA Hopper Architecture In-Depth. Nvidia · 22 March 2022
developer.nvidia.com/blog/nvidia-hopper-architecture-in-depth/

17. A Foveros Direct 3D Technology Brief. Intel · 6 February 2026
www.intel.com/content/dam/www/central-libraries/us/en/documents/2025-11/foveros-direct-3d-tech-brief.pdf

18. A [Tech Note] Hybrid Bonding: Evolving into a Foundational Technology for Improving Semiconductor Performance. SK hynix · 25 August 2026
news.skhynix.com/en/tech-note-series-ep2/

19. A TSMC, CoWoS technology page. TSMC · Accessed 9 September 2026
3dfabric.tsmc.com/english/dedicatedFoundry/technology/cowos.htm

20. A TSMC 2025 Annual Report, chapter 5. TSMC · 16 April 2026 · p. 3
investor.tsmc.com/static/annualReports/2025/english/pdf/2025_tsmc_ar_e_ch5.pdf#page=3

21. B Nvidia's Blackwell Reworked - Shipment Delays & GB200A Reworked Platforms. SemiAnalysis · 4 August 2024
semianalysis.com/2024/08/04/nvidias-blackwell-reworked-shipment/

22. B Hybrid Bonding Process Flow - Advanced Packaging Part 5. SemiAnalysis · 9 February 2024
newsletter.semianalysis.com/p/hybrid-bonding-process-flow-advanced

23. A TSMC 2022 Annual Report, page 97. TSMC · Accessed 18 October 2026
investor.tsmc.com/static/annualReports/2022/english/ebook/files/basic-html/page97.html

24. A Q2 2024 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese, English) — edited transcript. Refinitiv StreetEvents (transcript of a TSMC earnings call), via TSMC · 18 July 2024 · p. 6
investor.tsmc.com/english/encrypt/files/encrypt_file/reports/2024-08/5122725a56670882d777a8e8bfe0ed247cc55330/TSMC%202Q24%20Transcript.pdf#page=6

25. A Q3 2025 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese, English) — edited transcript. Refinitiv StreetEvents (transcript of a TSMC earnings call), via TSMC · 16 October 2025 · p. 7
investor.tsmc.com/english/encrypt/files/encrypt_file/reports/2025-10/6860312f04fd291d0f26b46c1234f84e6332717e/TSMC%203Q25%20Transcript.pdf#page=7

26. A Q2 2026 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese, English) — edited transcript. LSEG StreetEvents (transcript of a TSMC earnings call), via TSMC · 16 July 2026 · p. 17
investor.tsmc.com/english/encrypt/files/encrypt_file/reports/2026-08/3e494f0c14dd0890f897aa044415e21d93486cc4/TSMC%202Q26%20Transcript.pdf#page=17

27. A Applied Materials, Kinex integrated die-to-wafer hybrid bonding system product page. Applied Materials · Accessed 9 September 2026
www.appliedmaterials.com/eu/en/product-library/kinex-integrated-die-to-wafer-hybrid-bonding-system.html

28. A Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) · Accessed 9 September 2026
chipexplorer.eto.tech/?parentNode=N69&selectedNode=N71

29. A TSMC Announces Additional $100 Billion Investment in Arizona. Arizona Commerce Authority · 16 July 2026
www.azcommerce.com/news-events/news/2026/7/tsmc-announcement/

30. A Amkor Technology, annual report on Form 10-K for the fiscal year ended 31 December 2025. U.S. Securities and Exchange Commission (filing by Amkor Technology) · 20 February 2026
www.sec.gov/Archives/edgar/data/1047127/000104712726000014/amkr-20251231.htm

31. A Q3 2025 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese, English) — edited transcript. Refinitiv StreetEvents (transcript of a TSMC earnings call), via TSMC · 16 October 2025 · p. 13
investor.tsmc.com/english/encrypt/files/encrypt_file/reports/2025-10/6860312f04fd291d0f26b46c1234f84e6332717e/TSMC%203Q25%20Transcript.pdf#page=13

32. A JCET Accelerates Strategic Shift Toward High-End Advanced Packaging, 2025 Advanced Packaging Revenue Hits Record High. JCET Group · 21 April 2026
www.jcetglobal.com/en/site/news-detail?id=2052

33. A TSMC 2025 Annual Report, chapter 5. TSMC · 16 April 2026 · p. 4
investor.tsmc.com/static/annualReports/2025/english/pdf/2025_tsmc_ar_e_ch5.pdf#page=4

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A company filings, governments, standards bodies, universities and research institutions; B SemiAnalysis, Reuters, the Financial Times and Nikkei Asia.

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